



GridSense AI

See your power. Spend less.

An energy-intelligence platform for Rwanda — homes · landlords · enterprises

<https://gridsense-ai-zeta.vercel.app>

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The problem

You can't manage what you can't see

- Oct 2025: residential power became steeply **tiered** — **89** → **310** → **369 RWF/kWh**. Average tariff up ~**15%**.
- Households pay via prepaid "**cash power**" — they see a falling balance, not which appliance drains it.
- No warning when a home crosses from a cheap tier into an expensive one.



Creep from 20 → 21 kWh and the marginal unit costs **3.5x more** — with zero visibility into why.

The insight that reshaped our strategy

We confront the brutal truth instead of hiding it

- ~**50%** of connected homes use $\leq$ **20 kWh/month** (the 89 RWF lifeline). A 20% saving there $\approx$ **356 RWF (~\$0.25)/mo**. No one buys a device for that.
- The money — and willingness to pay — is with homes, landlords & SMEs that **cross into the 310 & 369 RWF tiers**.
- So we don't target "every household." We target the tier-crossers.

The solution

- A low-cost kit: **whole-home CT clamp + ESP32 (Wi-Fi)** for real-time total use & live RWF cost, **plus energy-metering smart plugs** for the biggest appliances.
- A simple **web app + tariff-aware AI**: live cost, bill forecast, **tier-cliff alerts**, and specific savings advice.
- Localized: **RWF, Rwanda's real tiers, Kinyarwanda-ready.**

One platform — three customers

Same verified tariff engine powers all three

- **Home (free):** a household sees its own live cost, tier-cliff alerts & personalized savings.
- **Landlord / Property:** manage many units & tenants — **exact per-tenant bills**, invoices, reminders, get paid on time.
- **Enterprise:** hotels & large compounds — many properties, teams, exports, connected devices.

Landlords: bill fairly, get paid on time

The feature that ends "I don't know how I'm charged"

- Add **properties** → **units** → **tenants** (name, phone, email) + meter readings.
- **Exact amount due** per tenant from the real tariff — with billing period and due date.
- One tap: **Invoice / Receipt / Consumption Report** (printable) → **send via WhatsApp / SMS / email**.
- **Payment reminders** as a calendar (.ics): 5 / 3 / 2 / 1 days before, the morning of, and 12:00 on the due day.
- Transparent, itemized charges that respect tenant privacy (Law 058/2021).



Plans & pricing

Individuals free for adoption; landlords & enterprise pay

Plan	Price (indicative)	For
Home	Free	A single household
Landlord — Starter	~RWF 9,900/mo (≤5 units)	Small landlords
Landlord — Growth	~RWF 29,000/mo (≤20 units)	Growing portfolios
Enterprise	from ~RWF 150,000/mo	Hotels, estates, teams
Device add-on	kit 40,100–69,100 + ~2,500/device/mo	Automatic live data

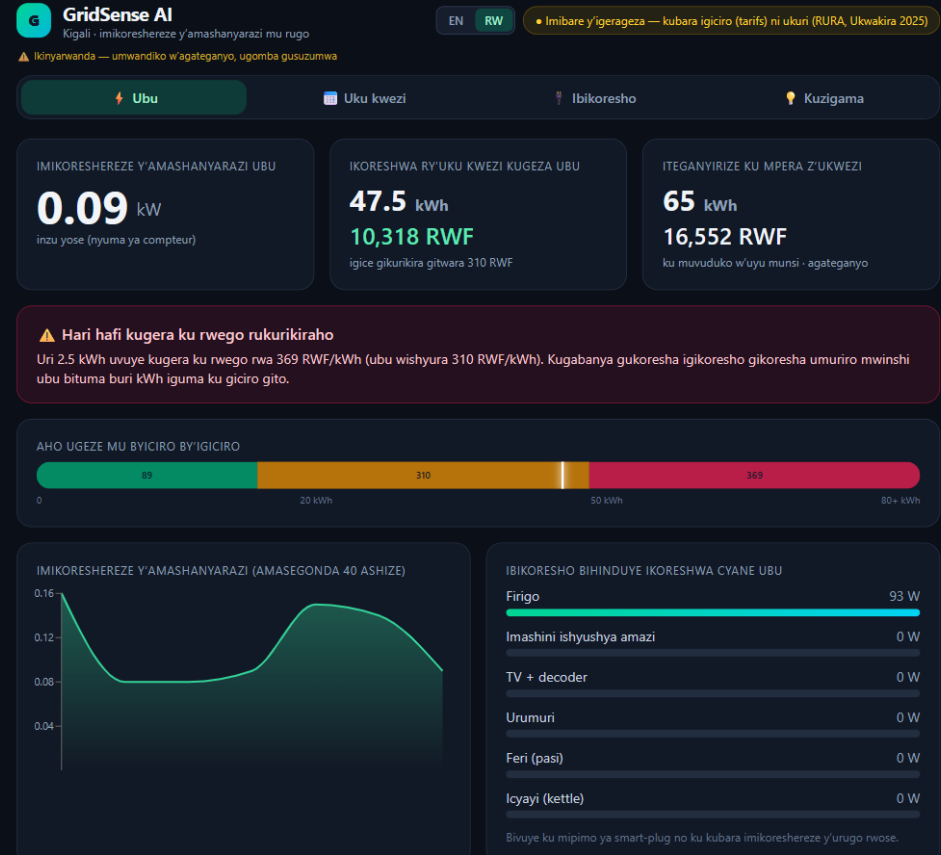
Premium checkout + cart are live; **demo mode charges no card** and is payment-provider-ready (MTN MoMo / Flutterwave / Stripe).

Live demo

English & Kinyarwanda — same app, one toggle



GridSense AI · Phase-1 demo · data source: `simulated` · tariff: 89 / 310 / 369 RWF/kWh (RURA, 1 Oct 2025)



GridSense AI · Igerageza rya Phase-1 · isoko: `simulated` · igiciro: 89 / 310 / 369 RWF/kWh (RURA, 1 Ukwakira 2025)

4 screens — Live Now · This Month · Appliances · Save. **Sensor data is simulated & labelled; the tariff math is real.**

Built & tested: 9 unit + 5 e2e tests · Lighthouse 100/100/100/100 · 05-build/

gridsense-ai-zeta.vercel.app

Is the "AI" real? — told honestly

- **Now:** tariff-aware analytics (kWh → live RWF across real tiers) + **rule-based** recommendations + run-rate **forecasting**.
- **Next:** ML bill forecasting + anomaly detection ("your fridge is drawing 30% more than usual").
- **Later:** appliance disaggregation (**NILM**) as data grows.
- **We never claim NILM we haven't built.**

How it works — and why it's buildable

- Non-invasive **CT clamp on the main feed**; **ESP32** reads current and uploads over Wi-Fi.
- Built on **proven open-source** designs (ESP32 + SCT-013 + EmonLib; CircuitSetup).



The simulated demo emits the **same JSON contract** — real hardware swaps in with a one-line change.

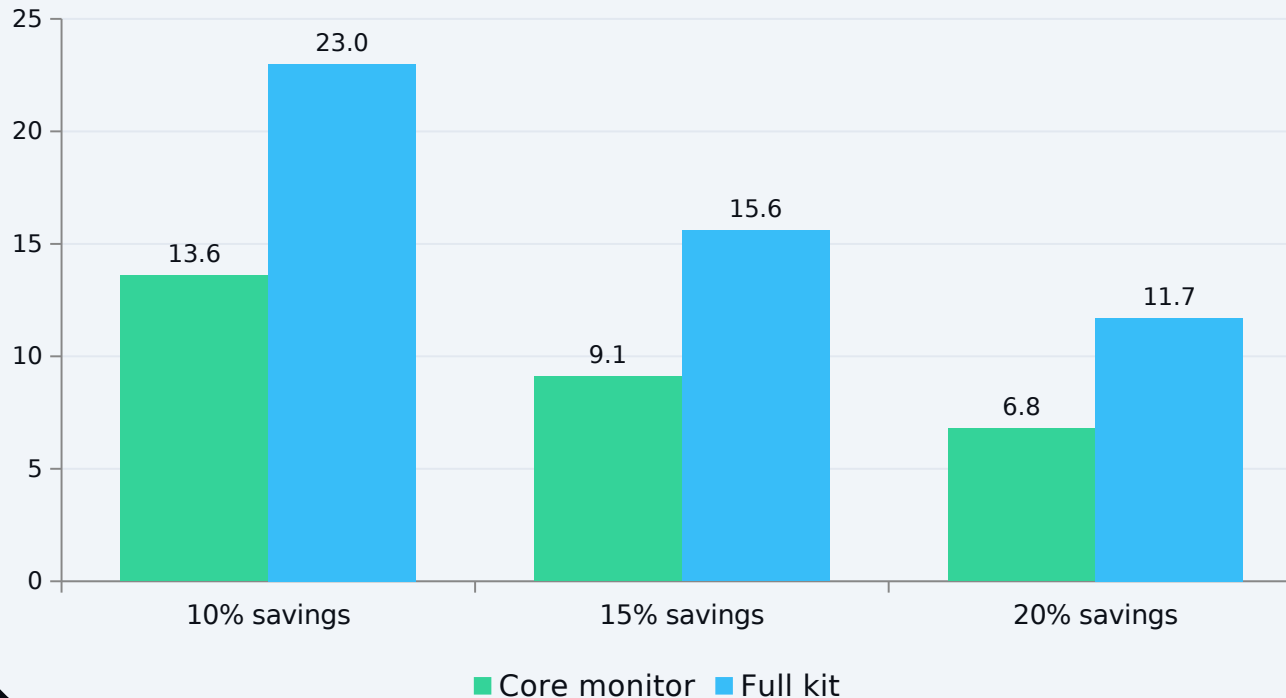
Feasibility & cost — sourced, in RWF

Component	Local price (RWF)	Source
ESP32 dev board	15,500	SoftTech Supply (Kigali)
SCT-013-100 CT clamp	12,000	SoftTech Supply
Metering smart plug	14,500	SoftTech Supply
PSU + wiring	7,600	SoftTech Supply
Whole-home monitor (entry SKU)	~40,100 (~\$27)	core unit
Full kit (+2 plugs)	~69,100 (~\$47)	1 USD ≈ 1,464 RWF

Unit economics & payback — the math is real

Verified tiers → a **100 kWh home pays 29,530 RWF/month**. Payback = kit cost ÷ monthly RWF saved.

Months to payback — 100 kWh home (lower is better)



- Core monitor: **~9 months @ 15%** savings.
- Heavy users (200 kWh): **~6 months @ 10%**.
- **< 12 months** for tier-crossers ✓
- Shown as a **range** — we don't fake one number.

Market & beachhead

- **1.946M** households on the grid; **84.6%** national electricity access.
- Beachhead = **tier-crossing urban homes + landlords/compounds + SMEs** — they cross 310/369, own appliances, have Wi-Fi + smartphones, and feel the hike.
- Mass-market & lifeline homes = a later **social-impact tier** (grant-funded), not the paying entry.

Competition & our moat

- Sense (~\$300, US, ML) · Emporia Vue · IoTaWatt (DIY) · Bidgely (utility B2B).
- **None** is localized: no RWF-tier logic, no Kinyarwanda, not Rwanda-priced or -serviced.
- **Moat = localization + tariff-aware AI + go-to-market**, at a Rwandan price.

Regulation & compliance — a credibility asset

- **Behind-the-meter, non-invasive** — no utility permit to clip a CT on your own wiring; **we never touch the REG/EUCL meter.**
- Board-side install by a **licensed electrician** (REG/RURA requirement); smart plugs are plug-and-play.
- **Data:** compliant with **Law 058/2021**; register with **NCSA**; consent + encryption + data minimization.

Go-to-market & funding — the lean ladder

- **\$0 to win the capstone** (free cloud, open-source, ALU resources). Funding is for scaling.
- Ladder: demo → **Hanga PitchFest** (50M RWF; RDB+MINICT+UNDP) → 3-5 kits + 1 pilot → larger grants (FEC 2026, Youth4Climate \$30k, ClimaFii \$70k, EEP Africa €200-500k) → production.
- Channel: direct + **landlord/SME** partnerships + possible REG partnership.

Roadmap

- **Phase 1 (now):** whole-home CT + ESP32 (Wi-Fi) + plugs; tariff-aware analytics, alerts, rules; web app. (demoed today)
- **Phase 2:** multi-circuit "by-area" CT in the board (partner electrician); GSM option; landlord sub-metering.
- **Phase 3:** ML forecasting → anomaly detection → NILM disaggregation.

Risks & honest mitigations

Risk	Mitigation
Savings % is an assumption (we show a 5-20% range)	Pilot will measure it; cite behavioral-feedback literature
Hardware funding (~\$27-47 / kit)	\$0 capstone path; bench PoC if free parts; Hanga/grants to scale
Wi-Fi penetration in target homes	Beachhead chosen for Wi-Fi/smartphone ownership; GSM in Phase 2
Hanga 2026 dates unconfirmed	Monitoring official channels; prototype-ready now

Impact & the ask

- **Impact:** energy literacy, lower bills, less waste, reduced peak demand, climate — real and fundable.
- **The ask:** capstone validation · a first pilot home/landlord · a path into Hanga PitchFest.

GridSense AI — see your power, spend less.

Appendix — Q&A defense

Rehearse these cold

If they ask...	Answer
Who actually pays?	Tier-crossers: urban homes, landlords, SMEs. Economics on slides 3, 9, 10.
Sense/Emporia exist.	None localized / RWF-tier-aware / Kinyarwanda / Rwanda-priced (11).
Is the AI real?	Tariff-aware + rules now; ML next; NILM later — no over-claim (6).
Did you build hardware?	Proven off-the-shelf; real ingestion contract; live demo (5, 7).
Will it touch the meter?	No — behind-the-meter, non-invasive; licensed electrician for board (12).
Privacy?	Law 058/2021; NCSA registration; consent + encryption (12).
How do you make money?	Device sale + optional subscription; landlord/SME tiers (13).
Where did the numbers come from?	Live Kigali retailers (RWF) + verified RURA tiers; savings as a range (8, 9).